

Roll No 2 5 C S E 1 0 4 9



National Institute of Technology Goa

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech. First Year

Minor Examination, October 2025

Course Name: Matrices and Advanced Calculus

Date: 05/10/2025

Duration: 1 Hour

Course Code: MA100

Time: 10.00 AM - 11.00 AM

Max. Marks: 25

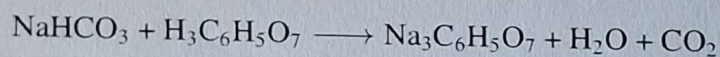
ANSWER ALL QUESTIONS

1. Solve using LU factorisation $2x + y + z = 5$, $3x + 5y + 2z = 15$, $2x + y + 4z = 8$ [5]

2. Express M as a linear combination of the matrices A, B, C , where $M = \begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$,
 $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix}$ [5]

3. Find basis and dimension of four fundamental bases of $A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 4 & 1 \\ 2 & -1 & 1 & 3 \end{bmatrix}$ [5]

4. Alka-Seltzer contains sodium bicarbonate (NaHCO_3) and citric acid ($\text{H}_3\text{C}_6\text{H}_5\text{O}_7$). When a tablet is dissolved in water, the following reaction produces sodium citrate, water, and carbon dioxide (gas):



Balance the chemical reaction equation using linear algebra methods by setting up a linear system of equations. [5]

5. Use only row operations to find the third column of A^{-1} , where $A = \begin{bmatrix} -2 & -7 & 9 \\ 2 & 5 & 6 \\ 1 & 3 & 4 \end{bmatrix}$ [5]

ALL THE BEST

Roll No 25CSE1049



National Institute of Technology Goa

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech. First Year

Mid Semester Examinations, October 2025

Course Name: Matrices and Advanced Calculus

Date: 14/10/2025

Duration: 1.5 Hours

Course Code: MA100

Time: 02.00 PM - 03.30 PM

Max. Marks: 50

ANSWER ALL QUESTIONS

1. Find the values of the parameter k for which the following system

$$x + y + z = 1,$$

$$2x + 3y + z = 2,$$

$$x + 2y + kz = 3,$$

$$3x + 5y + (k+1)z = 5$$

has (a) a unique solution, (b) infinitely many solutions, and (c) no solution. [5]

2. Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & b \end{bmatrix}$$

Determine the value(s) of the scalar b for which the rank of A is 2 [5]

3. Let

$$B = \begin{bmatrix} 3 & 1 & 2 & 0 \\ 1 & 2 & 0 & 1 \\ 4 & 0 & 1 & 2 \\ 0 & 1 & 3 & 1 \end{bmatrix}$$

Find only the second and fourth columns of B^{-1} . [5]

4. Forward elimination changes $AX = b$ to a row reduced $RX = d$ and the complete solution is

$$C_1 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + C_2 \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}$$

a) What is the 3 by 3 reduced row echelon matrix R and what is d ?

- (b) If the process of elimination subtracted 3 times row 1 from row 2 and then 5 times row 1 from row 3. What matrix connects R and d to the original matrix A and b . Use this matrix to find A and b . [5]

5. Prove or disprove: The following are vector spaces over $\mathbb{R}$, where $V = \mathbb{R}^2$ and addition is defined as in $\mathbb{R}^2$. For $x = (x_1, x_2) \in V$ and $\alpha \in \mathbb{R}$, scalar multiplication is defined as:

(a) $\alpha x := (x_1, 0)$. (b) $\alpha x := (\alpha x_1, -\alpha x_2)$. [5]

6. Use the Gram-Schmidt process to find an orthonormal basis for the subspace of $\mathbb{R}^3$ spanned by

$$v_1 = (1, 1, 0), \quad v_2 = (1, 0, 1) \quad v_3 = (2, -1, 3).$$

7. Consider the matrix [5]

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

Find matrices Q and R such that $A = QR$. [5]

8. Find the projection of $b = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$ onto the subspace spanned by $a_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $a_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. [5]

9. Fit a parabola $y = a + bx + cx^2$ to the points $(1, 6)$, $(2, 11)$, $(3, 18)$. [5]

10. Suppose the matrix is given by

$$A = \begin{bmatrix} 0 & 1 & 2 & 2 \\ 0 & 3 & 8 & 7 \\ 0 & 0 & 4 & 2 \end{bmatrix}.$$

- (a) Find all the special solutions to $AX = 0$ and describe in words the whole nullspace of A .
 (b) Describe the column space of this particular matrix A . "All combinations of the four columns" is not a sufficient answer.
 (c) Find the basis and dimension for the column space.
 (d) Find the rank of A .
 (e) What is the reduced row echelon form $R^* = \text{rref}(B)$ when B is the 6 by 8 matrix $B = \begin{bmatrix} A & A \\ A & A \end{bmatrix}$ using the same matrix A . [5]

Roll No	2	5	C	S	E	1	0	4	9
---------	---	---	---	---	---	---	---	---	---



National Institute of Technology Goa

Cuncolim, Salcete, South Goa-403703

Programme Name: B.Tech.(First Year)

End Semester Examinations, December 2025

Course Name: **Matrices and Advanced Calculus**

Course Code: **MA100**

Date: 17/12/2025

Time: 10.00 AM - 01.00 PM

Duration: 3 Hours

Max. Marks: 100

ANSWER ALL QUESTIONS

PART A - MATRICES

1. Check the consistency and, if possible, solve the system of equations $Ax = B$, where

$$A = \begin{pmatrix} 2 & 1 & 4 & 8 \\ -1 & 1 & -2 & -1 \\ 2 & 3 & 4 & 6 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 \\ 0 \\ 6 \end{pmatrix}, \text{ using the Gauss elimination and LU decomposition methods, and compare your answers.} \quad [8]$$

2. (a) Diagonalise $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ -4 & 4 & 3 \end{bmatrix}$ and hence find A^{10} . [4]

- (b) Suppose x is an eigenvector of A corresponding to an eigenvalue λ . [4]

i. Show that x is an eigenvector of $5I - A$. What is the corresponding eigenvalue?

ii. Show that x is an eigenvector of $5I - 3A + A^2$. What is the corresponding eigenvalue?

3. If $V = \{(x, y, z) : x = y\}$, then show that V is a vector space under standard operations and find its basis and dimension. [8]

4. Find the complete solution $x = x_p + x_n$ by forward elimination on $[A|b]$: [8]

$$\begin{bmatrix} 1 & 3 & 2 & 1 \\ 2 & 6 & 5 & 3 \\ 3 & 9 & 8 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \\ 17 \end{bmatrix}.$$

5. Apply the Gram-Schmidt process to the columns of A given by $a_1 = (2, 2, 0)$, $a_2 = \left(\frac{1}{2}, 0, \frac{1}{2}\right)$, $a_3 = (0, 3, 3)$ to find out an orthonormal set q_1, q_2, q_3 . If $Q = [q_1 \ q_2 \ q_3]$, then find R such that $A = QR$. [8]

PART B - ADVANCED CALCULUS

6. Find the volume of the solid generated by revolving the "triangular" region bounded by the curve $y = \frac{4}{x^3}$ and the lines $x = 1$ and $y = \frac{1}{2}$ about

(a) the x -axis (b) the y -axis (c) the line $x = 2$ (d) the line $y = 4$. [8]

7. Find the absolute maximum and minimum values of $f(x, y) = x^2 - y^2 - 2x + 4y + 1$ on the region R : the rectangular region in the first quadrant bounded by the coordinate axes and the lines $x = 4$ and $y = 2$. [8]

8. (a) For each of the following problems, you are given a function $f(x)$, a point, and a positive number ϵ . If the limit $\lim_{x \rightarrow a} f(x)$ exists, determine a value $\delta > 0$ such that the formal ϵ - δ definition of the limit is satisfied. If the limit does not exist, provide a clear explanation for why no such δ can be found. [4]

i. $f(x, y) = -\frac{x}{\sqrt{x^2 + y^2}}$, $(x_0, y_0) = (0, 0)$, $\epsilon = 0.05$

ii. $f(x, y) = \frac{x + y}{x^2 + 1}$, $(x_0, y_0) = (0, 0)$, $\epsilon = 0.01$

- (b) Find each of the following limits, or explain that the limit does not exist. [4]

i. $f(x) = (\sqrt{x^2 + 3x} - \sqrt{x^2 - 2x})$ as $x \rightarrow \infty$

ii. $f(x, y) = -\frac{x}{\sqrt{x^2 + y^2}}$ as $(x, y) \rightarrow (0, 0)$

9. The voltage V in a circuit that satisfies the law $V = IR$ is slowly dropping as the battery wears out. At the same time, the resistance R is increasing as the resistor heats up. Find how the current is changing at the instant when $R = 600$ ohms, $I = 0.04$ amp $dR/dt = 0.5$ ohm/sec, and $dV/dt = -0.01$ volt/sec. [8]

10. Find the maximum value that $f(x, y, z) = x^2 + 2y - z^2$ can have on the line of intersection of the planes $2x - y = 0$ and $y + z = 0$. [8]

PART C - SERIES

11. Which of the following series converge absolutely, which converge conditionally, and which diverge?

Give reasons for your answers. [10]

(a) $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ (b) $\sum_{n=1}^{\infty} \frac{(-1)^n}{\ln(n+1)}$ (c) $\sum_{n=1}^{\infty} \frac{n!}{10^n}$ (d) $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$

12. Find the Fourier series representation of [10]

$$f(x) = \begin{cases} \sin 2x, & -\pi < x < \frac{-\pi}{2} \\ 0, & \frac{-\pi}{2} < x \leq 0 \\ \sin 2x, & 0 < x \leq \pi. \end{cases}$$

ALL THE BEST